Model

1 Model

1.1 Environment

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \ldots, n\}$. These locations are partitioned into municipalities $g \in G$:

$$S = \bigcup_{g \in G} S^g, \qquad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed supply of land $\bar{H}_i$, allocated between residential and commercial use by local zoning:

$$\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where $\bar{H}_i^R$ is residential floor space capacity and $\bar{H}_i^F$ is commercial floor space capacity. Both are chosen by the municipal government and assumed to be fully utilized in equilibrium.

1.1.1 Transportation

Locations are connected by a transportation network (S, E), where $E \subseteq S \times S$ is a set of directed edges (links). Transportation investment $I_{k\ell}$ is chosen at the *edge* level, for each $(k, \ell) \in E$ The utility cost of traveling through edge (k, ℓ) is

$$d_{k\ell} = exp(\kappa \times time_{k\ell}),$$

where $time_{k\ell}$ is the travel time on edge (k, ℓ) and $\kappa > 0$ is a scale parameter, following Ahlfeldt et al., 2015. The travel time $time_{k\ell}$ is a function of the edge-level traffic $n_{k\ell}$ and investment $I_{k\ell}$:

$$time_{k\ell} = \delta_{kl}^0 + \delta_{kl}^1 \frac{n_{k\ell}^\rho}{I_{k\ell}^\beta}.$$

Following Fajgelbaum and Schaal, 2020, δ_{kl}^0 is the free-flow travel time on edge (k, ℓ), $\delta_{kl}^1 > 0$ is a scale parameter, and $\rho, \beta > 0$ are elasticities of travel time with respect to traffic and investment respectively. Therefore for a given route r, the commuting cost τ_{ij} between residence i and workplace j is the product of the costs of all edges along that route:

$$\tau_{ijr} = \prod_{(k,\ell) \in r_{ij}} d_{k\ell}(I_{k\ell}, n_{k\ell}),$$

where r_{ij} is the route chosen by commuters from i to j.

There are three types of agents: households, competitive firms, and governments. Households choose residence and workplace locations. Firms produce a freely tradable good using labor and commercial floor space. Government decisions operate at two levels: municipal governments choose local zoning, and a metropolitan transportation authority chooses investment across all links. In the decentralized equilibrium, municipalities and the transportation authority choose policies simultaneously, each treating the other's decisions as given.

1.2 Households

A household chooses a residence location i, a workplace j, and a route r from a finite, exogenously given choice set R_{ij} of feasible routes between i and j (e.g. the k shortest GIS paths, or distinct road/transit alternatives) to maximize utility. Preferences over the numaire good Q_{ij} and residential floor space h^R_{ij} are Cobb-Douglas:

$$u_{ijr}(\omega) = \frac{B_i}{\tau_{ijr}} \left(\frac{Q_{ij}}{\alpha}\right)^{\alpha} \left(\frac{h^R_{ij}}{1-\alpha}\right)^{1-\alpha} v_{ij,r}(\omega),$$

where $B_i > 0$ is a residential amenity at location i, $\alpha \in (0,1)$ is the expenditure share on the composite good, τ_{ijr} is the commuting cost along route r, and $\{v_{ij,r}(\omega)\}$ is a vector of idiosyncratic shocks, jointly distributed according to a nested (generalized extreme value) Fréchet copula with cumulative distribution function

$$F(\{v_{ij,r}\}) = \exp\left[-\sum_{ij\in S^2} T_i E_j \left(\sum_{r\in R_{ij}} v_{ij,r}^{-\theta}\right)^{\epsilon/\theta}\right], \qquad \theta \geq \epsilon > 1. \tag{1}$$

The outer exponent is $+\epsilon/\theta$, not $-\epsilon/\theta$: a valid CDF must satisfy $F \to 1$ as any $v_{ij,r} \to \infty$ holding the rest fixed, which pins the sign down. T_i, E_j enter at the outer (i,j) nest only, not inside the inner route sum: they are location-pair accessibility scales, not route-specific ones, exactly as in the single-route model. Setting $R_{ij} = \{r\}$ for every (i,j) collapses (1) to the original univariate Fréchet marginal $\Pr(v_{ij} \leq z) = \exp(-T_i E_j z^{-\epsilon})$, confirming the nesting strictly generalizes it; $\theta \geq \epsilon$ is required for (1) to be a valid (positively correlated within-nest) GEV copula.

The household's budget constraint is

$$Q_{ij} + q^R_i h^R_{ij} \leq w_j - \phi(N_i, L_i),$$

where q^R_i is the residential rent at location i, w_j is the wage at workplace j. Let $\tilde{W}_{ij} = w_j - \phi(N_i, L_i)$ denote the household's income, where $\phi(N_i, L_i)$ is the utility cost from local congestion. Optimal demands are $Q^*_{ij} = \alpha\tilde{W}_{ij}$ and $h^{R*}_{ij} = (1-\alpha)\tilde{W}_{ij}/q^R_i$, giving the deterministic component of indirect utility for a given route,

$$\tilde{V}_{ijr} = \frac{B_i\,\tilde{W}_{ij}}{\tau_{ijr}\,(q^R_i)^{1-\alpha}}.$$

Upper nest choice : applying the GEV choice-probability formula to (1), households compare residence-workplace pairs using the route-choice-corrected value $\tilde{V}_{ij}$, with T_i, E_j

entering exactly as in the single-route model:

$$\pi_{ij} = \frac{T_i E_j \tilde{V}_{ij}^{\epsilon}}{\sum_{i'j' \in S^2} T_{i'} E_{j'} \tilde{V}_{i'j'}^{\epsilon}} = \frac{T_i E_j \tilde{V}_{ij}^{\epsilon}}{\Phi}, \tag{2}$$

where $\tilde{V}_{ij} = \dfrac{B_i \tilde{W}_{ij}}{\tau_{ij}(q_i^R)^{1-\alpha}}$ and $\tau_{ij} = \left(\sum_{r \in R_{ij}} \tau_{ijr}^{-\theta}\right)^{-1/\theta}$ is the Fréchet mean of route costs.such that the population choosing i,j pair is:

$$N_{ij} = T_i E_j \tilde{V}_{ij}^{\epsilon} \frac{N}{\Phi}, \tag{3}$$

Lower nest choice : conditional on choosing (i, j), the household chooses a route $r \in R_{ij}$. Following Allen and Arkolakis (2022), we can rewrite $\tau_{ij} = \left(\sum_{r \in R_{ij}} \tau_{ijr}^{-\theta}\right)^{-1/\theta}$ such that

$$\tau = ((I - A)^{-1})^{-1/\theta}, \tag{4}$$

where A is a matrix with elements (k,l) is $d_{kl}^{-\theta}$,such that for matrix τ, the element (i,j) is τ_{ij}. Accordingly, we can derive the link intensity as follows:

$$\pi_{jk}^{k\ell} = (\frac{\tau_{ij}}{\tau_{ik} d_{kl} \tau_{lj}})^{\theta} \tag{5}$$

Therefore I can derive the traffic on edge (k,l) as follows:

$$n_{k\ell} = \sum_{ij \in S^2} N_{ij} \pi_{jk}^{k\ell} \tag{6}$$

1.3 Firms

At each location j, competitive firms produce a freely tradable good using labor L_j and commercial floor space H_j^F:

$$Y_j = A_j L_j^{\gamma} (H_j^F)^{1-\gamma},$$

where $A_j > 0$ is location-specific productivity and $\gamma \in (0, 1)$. Taking wages w_j and commercial rents q_j^F as given, profit maximization yields:

$$L_j = (\frac{\gamma A_j}{w_j})^{1/(1-\gamma)} H_j^F, \qquad q_j^F = (1-\gamma)(\frac{\gamma}{w_j})^{\gamma/(1-\gamma)} A_j^{1/(1-\gamma)}.$$

1.4 Municipal Zoning

Land at each location competes for residential and commercial use. Each municipality $g \in G$ chooses an iceberg cost $\{t_i^F\}_{i \in S^g} \geq 0$ on commercial land use at each location: procedural burden — discretionary review, permitting delay, litigation risk, compliance — melts away a fraction of commercial floor space's usable value before it reaches the market. Because the same underlying land parcel can be put to either use, competitive landowners allocate it to whichever use pays the higher effective return; if both uses are active in equilibrium, this competition equalizes returns at the margin:

$$q_i^R = \frac{q_i^F}{1+t_i^F}. \tag{7}$$

Combined with euqation **??**, we can get

$$(1-\gamma)(\frac{\gamma}{w_i})^{\gamma/(1-\gamma)} A_i^{1/(1-\gamma)} = (1+t_i^F)\frac{(1-\alpha)\sum_j N\pi_{ij}\,\tilde{W}_{ij}}{\bar{H}_i^R}. \tag{8}$$

The municipality sets $\{t_i^F\}$ to maximize the total welfare of its residents. In line with the political economy of local land use (Mast, 2024; Rollet and Weiwu, 2025), the municipal government, acting as a representative of its residents, trades off (1) lower rents and reduced worker-congestion from shrinking commercial land — both favor a higher iceberg cost — against (2) the residential crowding that comes with the resulting in-migration, and (3) the loss of local wage income to residents who both live and work at i, both of which favor a lower iceberg cost.

$$\max_{\{t_i^F\}_{i\in S^g}} W^g = \sum_{i\in S^g} N_i\,\bar{U}_i \quad \text{s.t.} \quad (7), \quad \bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where N_i is the equilibrium population at location i and $\bar{U}_i = \Gamma\big(1-\frac{1}{\epsilon}\big)\,\Phi_i^{1/\epsilon}$ is the expected utility of a representative resident at location i, taking other municipalities' iceberg costs $\{t_{i'}^F\}_{i'\notin S^g}$ and transportation investment $\{I_{k\ell}\}$ as given. $\phi(N_i, L_i)$ is a disutility from residing at i, capturing two sources of local negative externalities: residential congestion from neighbors and the presence of workers near the residential location:

$$\frac{\partial\phi}{\partial N_i} > 0, \qquad \frac{\partial\phi}{\partial L_i} > 0,$$

where N_i denotes residential population and L_i denotes employment at location i (defined above). I parametrize the disutility in Cobb-Douglas form:

$$\phi(N_i, L_i) = \eta\, N_i^a\, L_i^b, \qquad \eta > 0,\ a, b > 0, \tag{9}$$

Since $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$, the chain rule reduces the municipal first-order condition to a single comparison,

$$\frac{dW^g}{dt_i^F} = \left(\frac{\partial W^g}{\partial \bar{H}_i^R} - \frac{\partial W^g}{\partial \bar{H}_i^F}\right) \tag{10}$$

$$\times\,\frac{d\bar{H}_i^R}{dt_i^F} = 0, \tag{11}$$

so the optimum reduces to signing $\partial W^g/\partial \bar{H}_i^R - \partial W^g/\partial \bar{H}_i^F$, given $d\bar{H}_i^R/dt_i^F \neq 0$ ((**??**) below).

Channels. Only the i-th term of $W^g = \sum_{i'\in S^g} N_{i'}\bar{U}_{i'}$ depends on $\bar{H}_i^R, \bar{H}_i^F$ to leading order, holding other locations' land use and the metro-wide index Φ fixed. Three exact partials of $\bar{U}_i = \Gamma(1-1/\epsilon)\Phi_i^{1/\epsilon}$ follow directly from $\Phi_i = \sum_j T_i E_j B_i^\epsilon (w_j - \phi_i)^\epsilon \tau_{ij}^{-\epsilon} (q_i^R)^{-\epsilon(1-\alpha)}$:

$$\frac{\partial \bar{U}_i}{\partial q_i^R} = -\frac{(1-\alpha)\bar{U}_i}{q_i^R}, \qquad \frac{\partial \bar{U}_i}{\partial \phi_i} < 0, \qquad \frac{\partial \bar{U}_i}{\partial w_i} = \frac{\pi_{ii|i}\,\bar{U}_i}{w_i - \phi_i} > 0,$$

where $\pi_{ii|i}$ is the conditional probability that a resident of i also works at i (the local live-work share): the first equality is exact since q_i^R enters Φ_i as a common factor across every j; the third holds because only the $j = i$ term of Φ_i depends on w_i, so the wage channel is scaled down by how many residents actually capture it.

Substituting these, together with $\phi_i = \eta N_i^a L_i^b$, into $\partial(N_i\bar{U}_i)/\partial\bar{H}_i^R - \partial(N_i\bar{U}_i)/\partial\bar{H}_i^F$ gives the explicit optimality condition

$$\underbrace{\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R}}_{(1)\text{ in-mover }\geq 0} + \underbrace{N_i \frac{\partial \bar{U}_i}{\partial q_i^R}\frac{\partial q_i^R}{\partial \bar{H}_i^R}}_{(2)\text{ rent }\geq 0} + \underbrace{N_i \frac{\partial \bar{U}_i}{\partial \phi_i}\,\eta a N_i^{a-1} L_i^b\,\frac{\partial N_i}{\partial \bar{H}_i^R}}_{(3)\text{ residential congestion }\leq 0}$$
$$- \underbrace{N_i \frac{\partial \bar{U}_i}{\partial \phi_i}\,\eta b N_i^a L_i^{b-1}\,\frac{\partial L_i}{\partial \bar{H}_i^F}}_{(4)\text{ worker-congestion relief }\geq 0} - \underbrace{N_i \frac{\partial \bar{U}_i}{\partial w_i}\frac{\partial w_i}{\partial \bar{H}_i^F}}_{(5)\text{ local-wage loss }\leq 0} = 0. \tag{12}$$

Terms (1)–(5) reproduce the trade-off: attracting residents and lowering rents (1)–(2), against the residential crowding those in-movers bring (3), net of the worker-congestion relief from shrinking commercial land (4), shrinking $\bar{H}_i^F$ contracts local labor demand along an upward-sloping labor supply curve — household workplace choice responds to w_i with finite elasticity ϵ (5). The municipality thus weighs the value of local jobs to its own live-work residents against the congestion those same jobs create.Given the monoticity of $H_i^R(t_i^F)$, the municipal first-order condition (10) on t_i^F is equivalent to the first-order condition on $\bar{H}_i^R$.

For now the total land supply is fixed and fully utilized. In a future extension, I will incorporate a endogenous land supply where developers choose whether or not to build.

1.5 Transportation Authority

The metropolitan transportation authority chooses investment $\{I_{k\ell}\}_{(k,\ell)\in E}$ to maximize aggregate household welfare, taking the full zoning vector $\{\bar{H}_i^R, \bar{H}_i^F\}$ and the route assignment $\{r_{ij}\}$ as given:

$$\max_{\{I_{k\ell}\}} N \cdot \Gamma\big(1 - \tfrac{1}{\epsilon}\big)\, \Phi^{1/\epsilon}$$

subject to the budget constraint

$$\sum_{(k,\ell)\in E} \delta_{k\ell}\, I_{k\ell} \leq K, \qquad I_{k\ell} \geq 0,$$

where $\delta_{k\ell}$ is the per-unit cost of investment on edge (k,ℓ) and K is the total infrastructure budget.

Let $\mu \geq 0$ be the Lagrange multiplier on the budget constraint. Because $\ln \tau_{ij} = \sum_{(k,\ell)\in r_{ij}} \ln d_{k\ell}$, investment on edge (k,ℓ) only affects the commuting cost of a residence-workplace pair (i,j) whose route uses that edge, through $\partial \ln \tau_{ij}/\partial I_{k\ell} = -\gamma/I_{k\ell}$ for $(k,\ell) \in r_{ij}$ (and 0 otherwise). Aggregating over all such pairs, the interior first-order condition for edge (k,ℓ) is

$$\frac{N\gamma}{I_{k\ell}}\,\Gamma\big(1 - \tfrac{1}{\epsilon}\big)\, \Phi^{1/\epsilon - 1} \sum_{(i,j)\,:\,(k,\ell)\in r_{ij}} T_i E_j\, \tilde{V}_{ij}^{\epsilon} = \mu\, \delta_{k\ell}. \tag{13}$$

The left-hand side is the marginal social benefit of investment on edge (k,ℓ): lower travel cost on that edge raises $\tilde{V}_{ij}$, and hence Φ, for *every* residence-workplace pair routed through it — not only pairs directly adjacent to k and ℓ. This is the sense in which investment spillovers arise endogenously from the network structure: an edge shared by many routes is more valuable to the authority than one used only by its own endpoints, holding the direct commuting-cost elasticity fixed. The authority equates this marginal benefit per dollar of investment across all active edges.

1.6 Equilibrium

Definition 1 (Decentralized Equilibrium)**.** A decentralized equilibrium is a profile

$$\left(\{I^*_{k\ell}\}_{(k,\ell)\in E},\ \{q_i^{R*},\ q_i^{F*},\ w_j^*\}_{i,j\in S}\right)$$

such that:

1. **Household optimality.** Commuting flows $\{N\pi^*_{ij}\}$ are given by the Fréchet probability formula at prices (q_i^{R*}, w_j^*) and investment I^*, with route-based commuting costs $\{\tau^*_{ij}\}$ and edge traffic $\{n^*_{k\ell}\}$ satisfying (6).

2. **Firm optimality.** Wages w_j^* and commercial rents q_j^{F*} satisfy the firm first-order conditions at $H_j^F = \bar{H}_j^{F*}$ and $L_j^* = N\sum_i \pi^*_{ij}$.

3. **Market clearing.** Residential floor space clears at q_i^{R*} given by (**??**), commercial floor space is fixed at $H_j^F = \bar{H}_j^{F*}$, and the labor market clears at $L_j^* = N\sum_i \pi^*_{ij}$.

4. **Municipal Nash equilibrium.** Each municipality g sets the iceberg cost $\{t_i^{F*}\}_{i\in S^g}$ — equivalently, induces $\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}_{i\in S^g}$ via (**??**) — to satisfy the first-order condition (10), given $\{\bar{H}_{i'}^{R*}, \bar{H}_{i'}^{F*}\}_{i'\notin S^g}$ and I^*; no municipality has a profitable deviation.

5. **Transportation authority optimality.** $\{I^*_{k\ell}\}$ satisfies the first-order condition (13) for every edge $(k,\ell) \in E$, maximizing aggregate welfare given $\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}$ subject to the budget constraint.

Conditions (iv) and (v) jointly constitute the simultaneous Nash equilibrium between municipalities and the transportation authority: each set of agents takes the other's choices as given and best-responds. The equilibrium iceberg cost $t_i^{F*} \equiv q_i^{F*}/q_i^{R*} - 1$ is computed from the equilibrium quantities in this profile, not solved for as a separate object.

1.7 Data and Calibration

Following Fajgelbaum and Schaal (2020)'s treatment of network and cost fundamentals, parameters split into three groups: estimated from the model's own equations (§Quantification), set from external estimates, and location/link fundamentals recovered as residuals or calibrated directly to observed infrastructure costs.

Data. Commuting flows $N\pi_{ij}$ and employment L_i: LODES/LEHD origin-destination data. Wages w_j and population N_i: ACS. Residential and commercial rents q_i^R, q_i^F: county appraisal district records or commercial listing data (CoStar), jointly identifying t_i^F via (7). Land area $\bar{H}_i$: parcel/GIS data. Network edges E, route assignment $\{r_{ij}\}$, and travel times, for $\delta^\tau_{k\ell}$: GIS road network shortest paths and observed transit line/transfer sequences. Historical investment $I_{k\ell}$ and per-unit construction costs $\delta_{k\ell}$: NTD and state DOT capital programs, at the edge level. Total budget K: observed metro-area transportation capital spending over the relevant period.

Estimated from model equations. ϵ, ρ, and the commuting-cost investment elasticity in (14); (η, a, b) from the congestion regression; t_i^F is not estimated at all, but read directly off rent data via (7).

Set externally. α, the consumption expenditure share, from the CEX or national accounts; γ, the firm's labor share in (**??**)'s underlying production function, from the local labor share of income (BEA). *Notation note:* the commuting-cost investment elasticity in $d_{k\ell} = \delta_{k\ell}^{\tau} n_{k\ell}^{\rho} / I_{k\ell}^{\gamma}$ currently reuses the symbol γ already assigned to the firm's labor share in $Y_j = A_j L_j^{\gamma} (H_j^F)^{1-\gamma}$ — these are distinct parameters with distinct empirical targets and should be given separate symbols before the calibration is finalized.

Fundamentals recovered as residuals. T_i, B_i (jointly, from the origin fixed effect) and E_j, A_j (from the destination fixed effect and the firm's first-order condition) so that the model exactly matches observed N_i, L_i, w_j, q_i^R at the calibrated parameters, following the model-inversion logic of Ahlfeldt et al. (2015) and Fajgelbaum and Schaal (2020).

References

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